

Spring Cloud

Communication entre les microservices

Objectifs :

1. Communication avec WebClient
2. Communication avec Spring Cloud OpenFeign

Communication avec WebClient

1. Ajouter l'attribut depCode à l'entité Teacher et à la classe TeacherDto

```
private String depCode;
```

2. Créer la classe DepartmentDto identique à celle du projet departemt-service

```
package com.teachers.dto;

import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

@Data
@NoArgsConstructor
@AllArgsConstructor
public class DepartmentDto {
    private Long id;
    private String DepName;
    private String DepCode;
}
```

3. Créer la classe APIResponseDto

```
package com.nadhemb.teachers.dto;

import lombok.AllArgsConstructor;
import lombok.Data;
import lombok.NoArgsConstructor;

@Data
@NoArgsConstructor
@AllArgsConstructor
public class APIResponseDto {
    private TeacherDto teacherDto;
    private DepartmentDto departmentDto;
}
```

4. Ajouter la dépendance webflux

```
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-webflux</artifactId>
</dependency>
```

5. Configurer WebClient comme Spring Bean

```
@SpringBootApplication
public class TeacherMicroserviceApplication {

    @Bean
    public WebClient webClient(){
        return WebClient.builder().build();
    }

    ...
}
```

6. Injecter WebClient dans la classe TeacherServiceImpl

```
@Service
@AllArgsConstructor
public class TeacherServiceImpl implements TeacherService {

    private TeacherRepository teacherRepository;

    private WebClient webClient;

    ...
}
```

7. Au niveau de la classe TeacherServiceImpl, modifier la méthode getTeacherById comme suit :

```
@Override
public APIResponseDto getTeacherById(Long id) {
    Teacher teacher = teacherRepository.findById(id).get();

    DepartmentDto departmentDto = webClient.get()
        .uri("http://localhost:8080/api/departments/" +
            teacher.getDepCode())
        .retrieve()
        .bodyToMono(DepartmentDto.class)
        .block();

    TeacherDto teacherDto = new TeacherDto(
        teacher.getId(),
        teacher.getFirstName(),
        teacher.getLastName(),
        teacher.getDepCode()
    );

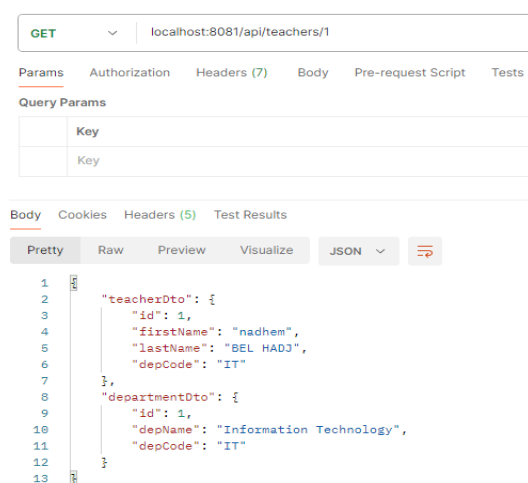
    APIResponseDto apiResponseDto = new APIResponseDto();
    apiResponseDto.setTeacherDto(teacherDto);
    apiResponseDto.setDepartmentDto(departmentDto);

    return apiResponseDto;
}
```

8. Au niveau du controller modifier la méthode getTeacherById comme suit :

```
@GetMapping("/{id}")
public ResponseEntity<APIResponseDto> getTeacherById(@PathVariable("id")
Long id )
{
    return new ResponseEntity<APIResponseDto>(teachersService.getTeacherById(id), HttpStatus.OK);
}
```

Tester votre travail avec Postman



Communication avec Spring Cloud OpenFeign

9. Ajouter la dépendance Spring Cloud OpenFeign

```
<dependency>
  <groupId>org.springframework.cloud</groupId>
  <artifactId>spring-cloud-starter-openfeign</artifactId>
</dependency>
```

10. Ajouter l'annotation @EnableFeignClients dans la classe principale TeacherMicroserviceApplication

```
@EnableFeignClients
public class TeacherMicroserviceApplication {
```

Créer un Feign API Client

11. Créer l'interface APIClient :

```
package com.nadhem.teacher.service;

import com.nadhem.teacher.dto.DepartmentDto;
import org.springframework.cloud.openfeign.FeignClient;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;

@FeignClient(url = "http://localhost:8080", value = "DEPARTMENT")
public interface APIClient {

    @GetMapping("api/departments/{department-code}")
    DepartmentDto getDepByCode(@PathVariable("department-code")
                               String departmentCode
    );

}
```

12. Modifier la méthode `getTeacherById` de la classe `TeacherServiceImpl`

```
public class TeacherServiceImpl implements TeacherService {

    @Autowired
    TeacherRepository teacherRepository;

    private APIClient apiClient;

    @Override
    public APIResponseDto getTeacherById(Long id) {
        Teacher teacher = teacherRepository.findById(id).get();

        DepartmentDto departmentDto = apiClient.getDepByCode(teacher.getDepCode());
        ...
    }
}
```

Tester votre travail avec Postman

Ajouter le nom du département à TeacherDto

la classe TeacherDto

```
public class TeacherDto {

    private Long id;
    private String firstName;
    private String lastName;
    private String depCode;
    private String DepName;
}
```

la classe TeacherServiceImpl

```
@Override
public APIResponseDto getTeacherById(Long id) {
    Teacher teacher = teacherRepository.findById(id).get();
    .....
}
```

```
TeacherDto teacherDto = new TeacherDto(
    teacher.getId(),
    teacher.getFirstName(),
    teacher.getLastName(),
    teacher.getDepCode(),
    departmentDto.getDepName()
);
```

```
1  {
2    "teacherDto": {
3      "id": 1,
4      "firstName": "nadhem",
5      "lastName": "BEL HADJ",
6      "depCode": "IT",
7      "depName": "Information Tech"
8    },
9    "departmentDto": {
10     "id": 1,
11     "depCode": "IT",
12     "depName": "Information Tech"
13   }
14 }
```